AI Innovation Hackathon Handbook

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Theme: Build an AI Co-Pilot for Industry

Overview

Teams will build an AI Co-Pilot that solves a real business problem using multimodal data, deep learning, explainable AI, and a Human-in-the-Loop workflow. The goal is to create a solution with commercial potential.

Mandatory Requirements

• Process at least three data modalities (images, text, PDFs, tabular data, time series, audio, etc.).

• Use at least one predictive deep learning model (ANN, CNN, or RNN/LSTM/GRU).

• LLMs may be used for reasoning, summarization, explanations, or report generation but not as the primary predictive engine.

• Include a Human-in-the-Loop workflow where a user approves, rejects, or modifies AI recommendations.

• Provide Explainable AI (confidence scores, feature importance, SHAP, Grad-CAM, etc.).

• Develop a working web application.

• Generate a downloadable PDF or DOCX report.

• Present a business model and commercialization strategy.

Suggested Industry Challenges

1. Insurance Claim Processing

Suggested datasets:

- Car Damage (Kaggle): https://www.kaggle.com/search?q=car+damage

- Roboflow Universe: https://universe.roboflow.com

- Insurance Claims (Kaggle): https://www.kaggle.com/search?q=insurance+claims

Suggested APIs:

- OpenWeather: https://openweathermap.org/api

- OpenStreetMap: https://nominatim.openstreetmap.org

2. Healthcare Assistant

Suggested datasets:

- Chest X-Ray Pneumonia: https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

- UCI Heart Disease: https://archive.ics.uci.edu/dataset/45/heart+disease

- PubMed Central: https://pmc.ncbi.nlm.nih.gov

Suggested APIs:

- OpenFDA: https://open.fda.gov

- NIH APIs: https://api.nih.gov

3. Agriculture Advisor

Suggested datasets:

- PlantVillage: https://www.kaggle.com/datasets/emmarex/plantdisease
- Soil datasets: https://www.kaggle.com/search?q=soil
- Crop datasets: https://data.worldbank.org

Suggested APIs:

- OpenWeather API: https://openweathermap.org/api

4. Manufacturing Quality Inspection

Suggested datasets:

- MVTec AD: https://www.mvtec.com/company/research/datasets/mvtec-ad
- Roboflow: https://universe.roboflow.com

Suggested APIs:

- OpenCV docs: https://opencv.org

5. HR Resume Screening

Suggested datasets:

- Resume Dataset: https://www.kaggle.com/search?q=resume
- Job Descriptions: https://huggingface.co/datasets

Suggested APIs:

- LinkedIn data not recommended; use public/synthetic data

6. Bank Loan Assistant

Suggested datasets:

- Loan Prediction: https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset
- UCI datasets: https://archive.ics.uci.edu/ml

Suggested APIs:

- ExchangeRate API: https://www.exchangerate-api.com

7. Retail Shelf Analytics

Suggested datasets:

- Open Images: https://storage.googleapis.com/openimages/web/index.html
- COCO: https://cocodataset.org

Suggested APIs:

- NewsAPI: https://newsapi.org

8. Education Analytics

Suggested datasets:

- Student Performance: https://archive.ics.uci.edu/ml/datasets/student+performance
- Exam datasets: https://www.kaggle.com/search?q=student+performance

Suggested APIs:

- None required

9. Supply Chain Risk

Suggested datasets:

- Supply Chain datasets: https://www.kaggle.com/search?q=supply+chain

Suggested APIs:

- OpenWeather
- OpenStreetMap

10. Smart City Road Damage

Suggested datasets:

- Road Damage Dataset: https://www.kaggle.com/search?q=road+damage

Suggested APIs:

- OpenStreetMap
- Geoapify: https://www.geoapify.com

General Dataset Sources

• https://www.kaggle.com/datasets

• https://archive.ics.uci.edu/ml

• https://huggingface.co/datasets

• https://datasetsearch.research.google.com

• https://www.openml.org

• https://data.worldbank.org

• https://registry.opendata.aws

Evaluation Rubric

• Business Problem – 15

• Multimodal Integration – 15

• Deep Learning Implementation – 15

• Human-in-the-Loop – 10

• Explainable AI – 10

• System Architecture – 10

• User Experience – 10

• Business Model – 10

• Innovation – 5

• Presentation – 10